

Meep for Electromagnetic Simulations

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This article focuses on the MIT electromagnetic equation propagation (Meep) software developed at Massachusetts Institute of Technology (MIT) for modelling electromagnetic (EM) systems. The implementation is based on finite-difference time-domain (FDTD) method. The latest release of this open source software released under GNU general public licence, version 1.3, is included in this month's EFY Plus.

FDTD simulation

In the FDTD numeric analysis technique, time and space are divided into a regular grid and the time evolution of Maxwell's equations is simulated. Of course, there are many numeric tools for computational electromagnetism. Some of these like Finite Element Methods (FEMs) are more sophisticated and guarantee greater accuracy.

Boundary Element Method represents interfaces instead of volume. Each technique has its own advantages and disadvantages.

Generic in nature. The full time-dependent Maxwell's equations can be directly applied for nearly arbitrary materials including anisotropic, dispersive, time varying and non-linear. The algorithm helps in realistic simulation and propagation of an EM wave through these complex structures. Also, the software is capable of testing a wide range of boundary conditions.

For researchers working in computational electromagnetism, new interactions of modern physical processes are easily possible with the help of an FDTD algorithm. An experienced user can simulate the population of excited atoms for lasing using this technique.

Distributed computing. The design of Meep facilitates distributed computing. The algorithm used in this software can be easily parallelised for

Software overview

Name of the software: Meep

Licence: GNU GPL

Software type: Electromagnetic simulation package

Major contributors to the software: David Roundy, Mihai Ibanescu, Peter Bermel, Steven G. Johnson and Ardavan F. Oskooi

Programming language: C++

System requirements: Any computer with Unix-like system and C++ compiler; optionally exploits additional free software packages like GNU Guile, libcti interface library, HDF5, MPI message-passing interface and Harminv filter-diagonalisation

For more information: ab-initio.mit.edu/meep

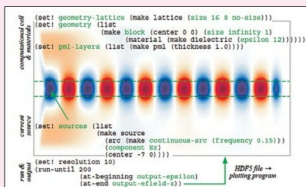


Fig. 1: Meep plot showing the field in a dielectric wave-guide excited by a point source at a given frequency. Input file in Scheme Language is shown at the foreground

running on clusters or supercomputers. Distributed memory parallelism could be achieved on any system that supports Message Passing Interfacing (MPI) standard. This simulation package is portable to any Unix-like system. With a free Unix-compatibility environment like Cygwin, it should be possible to install Meep in Windows too.

The motivation behind

There are plenty of commercial software packages based on FDTD algorithm. For any kind of improved scientific research in this area, access to the source code is essential. Licensing constraints should be relaxed in order to speed porting to new clusters and supercomputers. Therefore most research groups working on photonics and computational electromagnetics end up having their own FDTD

implementation. This results in duplication of effort and wastage of time and money.

There were a few FDTD implementations available to the public when the research group at MIT started working on Meep. But the features were limited and were not capable enough to meet their requirements. Massive down-

loads and citations of this package reaffirm the demand for such a software package.

Time-domain simulation: Major applications

In a time-domain simulation, Maxwell's equations are evolved over time in a finite computational region. The result of this numerical experiment is helpful to compute many useful quantities like the ones mentioned below:

By taking Fourier transform of the response to a short pulse, transmission and reflection-scattering parameters over a wide spectrum could be easily found out.

Frequencies, decay rates and field patterns of the harmonic modes of a system could be easily extracted from the analysis of the system to a short pulse.

Field patterns like Green's function